

Definition. A *cell product function* is a real valued function whose domain is the set of all binary lattices and whose value on any lattice is the product of its values on every cell in the lattice.

[[I'm proposing what's below as a theorem rather than proposition, to be parallel to Theorem 3. I feel both should be theorems or both propositions, and Theorem 3 seems worthy of being a theorem rather than a proposition.]]

Theorem 1. Let v be the cell product function that maps the cells $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, and maps the fourteen other cells to 1. Then for all positive integers m and n , the number of (m, n) polygon mosaics equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .

Proof. For any framed lattice ℓ , $v(\ell)$ is 1 if ℓ is proper, i.e., has neither $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ nor $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells, and otherwise is 0, so that $\sum_{\ell} v(\ell)$ counts the number of proper (m, n) framed lattices. Our conclusion now follows from the fact that $\mathcal{L}$ gives a bijection from the set of (m, n) polygon mosaics to the set of proper (m, n) framed lattices. $\square$

[[Here is a proposed rephrasing of material that on April 8 was in Section 4. For a major result like this, I think it might be better to put everything in one statement rather than spreading it through prior definitions.]]

Theorem 2. Let v be any cell product function. Define the 1×1 matrices $A_1 = \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$, $B_1 = \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \end{pmatrix}$, $C_1 = \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \end{pmatrix}$, and $D_1 = \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \end{pmatrix}$, and for positive integers n , define the $2^n \times 2^n$ matrices [[We used to say $2^{n-1} \times 2^{n-1}$; this bothered me for a while, and now I realize it was wrong!! The issue is that while A_n, B_n , etc. are indeed $2^{n-1} \times 2^{n-1}$, here we are defining A_{n+1}, B_{n+1} , etc.]]

$$\begin{aligned} A_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} D_n \end{pmatrix} & B_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} D_n \end{pmatrix} \\ C_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} D_n \end{pmatrix} & D_{n+1} &= \begin{pmatrix} v \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} A_n & v \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} B_n \\ v \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} C_n & v \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} D_n \end{pmatrix}. \end{aligned}$$

Then for all positive integers m and n , the upper left entry of A_n^m equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ . [[I changed “(0,0) entry” to “upper left entry” because it means a casual reader doesn't need to know how we're numbering matrix entries to understand this theorem's conclusion. Thoughts?]]

1. COUNTING POLYGON-FREE MOSAICS

Theorem 3. Let v be the cell product function that maps the cells $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to 0, $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ to 7, $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ and $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ to -1 , and the ten other cells to 1. Then for all positive integers m and n , the number of (m, n) polygon-free mosaics equals $\sum_{\ell} v(\ell)$, where the sum is over all (m, n) framed lattices ℓ .

Throughout this section, let v be as in Theorem 3. Thus, for example, if ℓ is the framed lattice on the right of Figure ??, we have $v(\ell) = -7^{13}$, as there are no $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells, twelve $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cells, one $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ cell, two $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cells, and one $\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}$ cell.

The following result, which is an important part of proving Theorem 3, connects local and global information in the sense that $v(\ell)$ is calculated by multiplying

values for each cell of ℓ , whereas the number of polygons is a characteristic of an entire mosaic.

Proposition 4. *For a polygon mosaic P , let $\ell = \mathcal{L}(P)$. If P contains an even [resp. odd] number of polygons then $v(\ell)$ is positive [resp. negative], where v is as in Theorem 3.*

Proof. [[Insert proof of this proposition here.]] □

Proof of Theorem 3. Fixing m and n , throughout this proof we use “polygon mosaic” for “ (m, n) polygon mosaic,” “framed lattice” for “ (m, n) framed lattice,” and so on.

We introduce the following identification for polygon mosaics. Let $\{P_i\}_{i \in J}$ be the set of all polygon mosaics that have exactly one polygon, and for any subset I of J , let P_I , if it exists, be the polygon mosaic that contains exactly the polygons of P_i for all $i \in I$. For example, $P_\emptyset$ is the polygon mosaic containing no polygons, i.e., the mosaic entirely composed of blank tiles, and for any $i \in J$, $P_{\{i\}}$ always exists and equals P_i . For an arbitrary $I \subseteq J$, P_I exists if and only if the polygons of P_i , for $i \in I$, do not “overlap.” More precisely, P_I exists if and only if for every mosaic location, there is at most one $i \in I$ such that P_i has a non-blank tile there. Notice that every polygon mosaic can be uniquely written as P_I for some $I \subseteq J$.

Next we look at some sets of mosaics. For any $I \subseteq J$ such that P_I exists, let M_I be the set of mosaics that contain the polygons of P_I , i.e., the set of mosaics that can be formed from P_I by using any tile wherever P_I has a blank tile. Notice that the size $|M_I|$ of this set equals 7^a , where a is the number of blank tiles in P_I . By the definition of $\mathcal{L}$, the number of cells of $\mathcal{L}(P_I)$ that are either $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ or $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ is also a , and since $\mathcal{L}(P_I)$ is proper, i.e. it has neither $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ nor $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ cells, we see that $|v(\mathcal{L}(P_I))| = |M_I|$. Furthermore, $|I|$ is the number of polygons in P_I , so Proposition 4 tells us that $v(\mathcal{L}(P_I))$ and $(-1)^{|I|}$ have the same sign. Therefore, if P_I exists then

$$v(\mathcal{L}(P_I)) = (-1)^{|I|} |M_I|.$$

We must show that the the sum $\sum_{\ell} v(\ell)$ over all framed lattices ℓ is equal to number of polygon-free mosaics. The sum is unchanged if we restrict to the sum over all proper framed lattices ℓ because $v(\ell) = 0$ if ℓ is not proper. Furthermore, since $\mathcal{L}$ gives a bijection from polygon mosaics to proper framed lattices,

$$\sum_{\ell} v(\ell) = \sum_I v(\mathcal{L}(P_I)) = \sum_I (-1)^{|I|} |M_I|,$$

where the second and third sums are over all $I \subseteq J$ such that P_I exists. The total number of mosaics, which happens to be 7^{mn} , equals $|M_\emptyset|$. Notice that if $J = \emptyset$ if and only if $m = 1$ or $n = 1$. If this is the case then every mosaic is polygon-free, the third sum above is $|M_\emptyset|$, and our theorem is complete. Therefore we may assume that $m \geq 2$ and $n \geq 2$, so that $J \neq \emptyset$ and the third sum has a nonzero term with $I \neq \emptyset$, for example, $I = \{i\}$ for any $i \in J$.

For any $I \neq \emptyset$ such that P_I exists, $M_I = \bigcap_{i \in I} M_{\{i\}}$ by the definition of the M_I . Thus

$$\sum_{\ell} v(\ell) = |M_\emptyset| - \sum_I (-1)^{|I|-1} \left| \bigcap_{i \in I} M_{\{i\}} \right|,$$

where the sum on the right is over all I such that P_I exists and $\emptyset \neq I \subseteq J$. In this equation, the value on the right side is unchanged if instead we sum over all I such that $\emptyset \neq I \subseteq J$. This is because when P_I is not defined, the P_i , $i \in I$, “overlap” in the sense described above, and so $\bigcap_{i \in I} M_{\{i\}} = \emptyset$. By the inclusion-exclusion principle (see the Appendix), this summation over all I with $\emptyset \neq I \subseteq J$ equals $\left| \bigcup_{i \in J} M_{\{i\}} \right|$, i.e., the number of mosaics that contain at least one polygon. Since $|M_\emptyset|$ is the total number of mosaics, we conclude, as desired, that $\sum_\ell v(\ell)$ is the number of polygon-free mosaics. $\square$